# Product Specification

Model Name: LCMDO4R97MZC1

Description：4.97〃 HD(720x1280) AMOLED

Doc. Version： 01

Customer： Common

□Approved for Preliminary Specification

□Approved for Final Specification

■Approved for Final Specification & Sample

| Prepared | Checked | Approved |
|---|---|---|
| | | |

Customer’s Approval

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|---|
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Version History

| Version. No | Date | Contents | Remark |
|---|---|---|---|
| 01 | 2022-09-19 | First Version | |
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# 1 Scope

This Specification defines AMOLED manufactured by Wisevast (Chengdu)ElectronicTechnology Co.,Ltd., from here on refer as WISEVAST. In the case of any unspecified item, it may require both WISEVAST and the party designs this module into its product to work out a solution.

# 2 Features

## 2.1 Product Applications

Mobile phone

## 2.2 Product Features

1) Display color：16.7M (RGB x 8bits)
2) Display format：4.97" HD (720RGBx1280)
3) Pixel arrangement：Real RGB arrangement
4) Interface：MIPI 4 lanes
5) State：W/ POL

# 3 Mechanical Specifications

| Item | Specification | unit | Note |
|---|---|---|---|
| Panel Dimension outline | 64.12 × 116.72×0.643 | mm | * |
| LTPS Glass outline | 64.12 × 116.72 | mm | |
| Encapsulation Glass outline | 64.12 × 113.82 | mm | |
| Number of dots | 720(W) x RGB x 1280(H) | dots | |
| Active area | 61.884 × 110.016 | mm | |
| Diagonal size | 4.97 | inch | |
| Pixel pitch | 28.65 × 85.95 | μm | |
| Glass thickness (LTPS/encapsulation glass) | 0.2 / 0.3 | mm | |
| Weight | TBD | g | |

*Note: Refer to 9 Outline Dimension Drawing

# 4 Maximum Rating

| Parameter | Symbol | Spec | | | Unit | Note |
|---|---|---|---|---|---|---|
| | | Min. | Typ. | Max. | | |
| Analog/boost power voltage | VCI | -0.3 | - | 5.5 | V | - |
| VCI I/O voltage | VCI_IF | -0.3 | - | 5.5 | V | - |
| I/O voltage | VDDI | -0.3 | - | 5.5 | V | - |
| VSP voltage | VSP | -0.3 | - | 6.6 | V | - |
| VPP(OTP power) | VPP | - | - | 8.25 | V | - |

# 5 Electrical Specifications

## 5.1 Electrical Characteristics

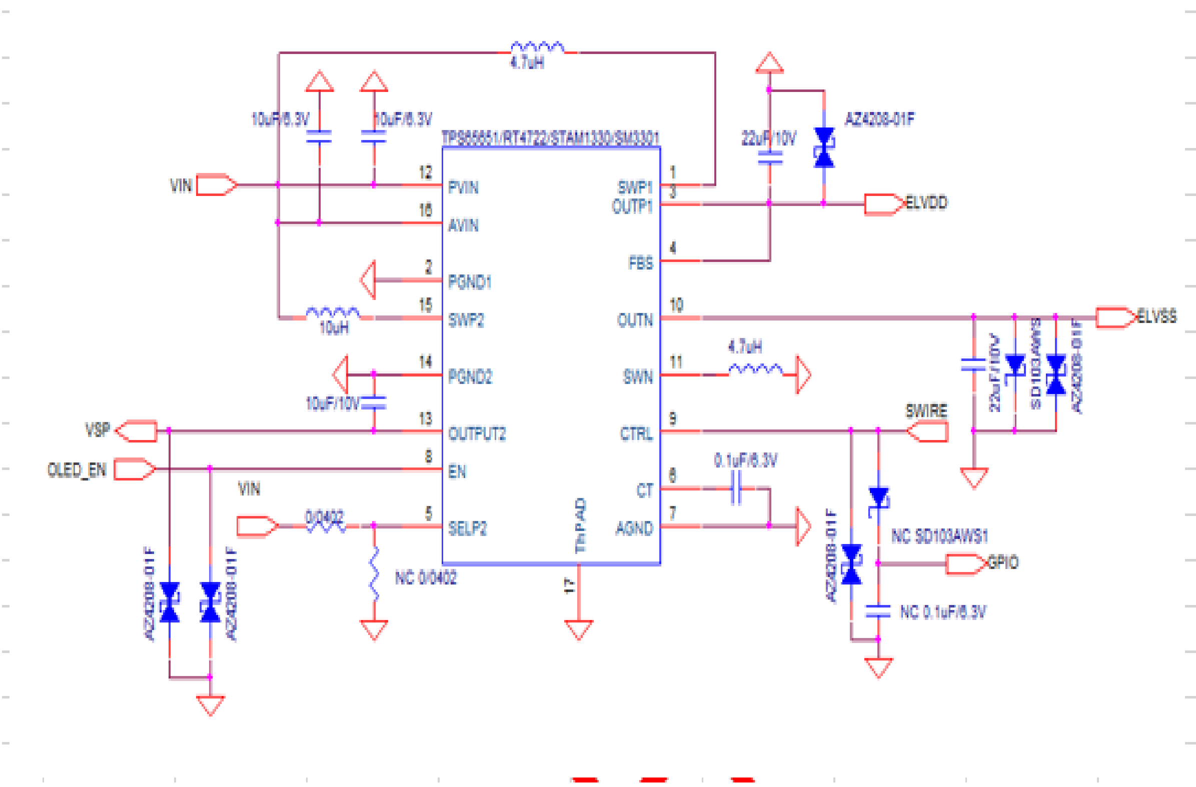
### 5.1.1 Power Characteristic

| Item | Symbol | Min. | Typ. | Max. | Unit | Remark |
|---|---|---|---|---|---|---|
| AMOLED Power positive | ELVDD | | 4.6 | | V | |
| AMOLED power Negative | ELVSS | | -2.5 | | V | Ref |
| Gamma Voltage | VSP | | 6.4 | - | V | Ref |
| Digital Power supply | VDDI | 1.65 | 1.8 | 3.3 | V | Ref |
| Analog Power supply | VCI | 2.5 | 3.3 | 3.6 | V | Ref |

<table>
<tr><th>Mode</th><th>Symbol</th><th>Condition</th><th>Min.</th><th>Typ.</th><th>Max.</th><th>Unit</th></tr>
<tr><td rowspan="4">350 nits<br>@Gray 255</td><td>$I_{ELVDD/ELVSS}$</td><td rowspan="4">VELVDD=4.6V<br>VELVSS=-2.5V<br>VCI=3.3V<br>VDDIO=1.8V<br>VSP=6.4V</td><td>-</td><td>160</td><td>190</td><td>mA</td></tr>
<tr><td>$I_{VCI}$</td><td>-</td><td>1.5</td><td>2</td><td>mA</td></tr>
<tr><td>$I_{VDDIO}$</td><td>-</td><td>15</td><td>20</td><td>mA</td></tr>
<tr><td>$I_{VSP}$</td><td>-</td><td>10</td><td>13</td><td>mA</td></tr>
</table>

## 5.1.2 Power supply circuit application (Reference only)

Power IC recommend: Silicon Mitus: SM3301



## 5.2 FPC Connection and Block Diagrams

### 5.2.1 Main FPC PIN Descriptton

| PIN | SYMBOL | Description |
|---|---|---|
| 1 | GND | 接地口 GND |
| 2 | VBAT | POWER 3.3V |
| 3 | VBAT | POWER 3.3V |
| 4 | VBAT | POWER 3.3V |
| 5 | VBAT | POWER 3.3V |
| 6 | VBAT | POWER 3.3V |
| 7 | GND | 接地口 GND |
| 8 | GND | 接地口 GND |
| 9 | GND | 接地口 GND |
| 10 | NC | 空脚 |
| 11 | TE | Output for monitoring |
| 12 | TRESX | LCD复位脚,低电平有效 |
| 13 | VDDI | Driver ic digital |
| 14 | GND | 接地口 GND |
| 15 | D2P | MIPI DSI data2+ |
| 16 | D2N | MIPI DSI data2- |
| 17 | GND | 接地口 GND |
| 18 | DIP | MIPI DSI data1+ |
| 19 | DIN | MIPI DSI data1- |
| 20 | GND | 接地口 GND |
| 21 | CKP | MIPI DSI clock+ |
| 22 | CKN | MIPI DSI clock- |
| 23 | GND | 接地口 GND |
| 24 | DOP | MIPI DSI data0+ |
| 25 | DON | MIPI DSI data0- |
| 26 | GND | 接地口 GND |
| 27 | D3P | MIPI DSI data3+ |
| 28 | D3N | MIPI DSI data3- |
| 29 | GND | 接地口 GND |
| 30 | VCI | driver ic analog supply 2.8V |
| 31 | GND | 接地口 GND |
| 32 | TP-VCC | Touch panel analog supply 3.3V |
| 33 | TP-VDDI | Touch panel I/O voltage supply 3.3V |
| 34 | TP-INT | Touch panel interrupt output |
| 35 | TPSDA | Touch panel I2C data |
| 36 | TPSCL | Toutch panel I2C clock |
| 37 | TPRESX | Touch panel reset |
| 38 | TPA4 | Touch panel reserved GPIO |
| 39 | GND | 接地口 GND |
| 40 | NC | 空脚 |

## 5.2.3 TP Sensor

| No | Item | Specification | Remark |
|---|---|---|---|
| 1 | Structure | On cell | |
| 3 | Pitch(A) | 4.0mm <A<4.6mm | Be suitable for TX and RX |
| 4 | Full capacitive limited Range | Avg +/- 6sigma | |
| 5 | Difference check of Base Line between each points | T(x) - T(x+1) < 390 fF<br>R(x) - R(x+1) < 390 fF | |
| 6 | Noise limited Range | -10≦X≦10 | |
| 8 | Quantity of valid ACF grain | More than 10 grains/each bonding pad Needed | |
| 9 | Sealing compound on Bonding area and touch IC | Invisible | |
| 10 | Obvious ITO etching pattern | | |

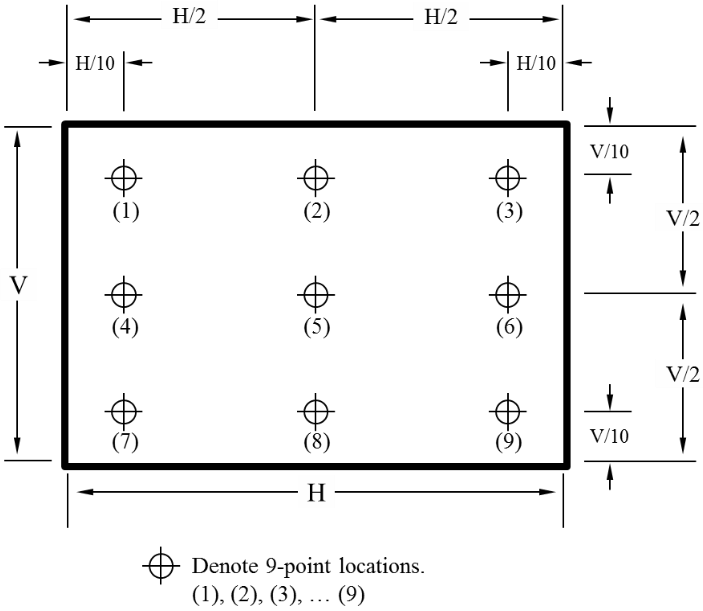
# 6 Electro-Optical Specification

| Item | | Symbol | Conditions | Min. | Typ. | Max. | Unit | Remark |
|---|---|---|---|---|---|---|---|---|
| Brightness | | | | 315 | 350 | 385 | cd/m2 | Suggest MDL Brightness |
| Brightness Uniformity | | | | 75 | 85 | - | % | Note 1 |
| Contrast Ratio | | CR | | 10000 | 20000 | | | Based on CA-310 Note 2 |
| CIE | Red | x | | 0.660 | 0.690 | 0.720 | - | Set F |
| | | y | | 0.281 | 0.311 | 0.341 | - | |
| | Green | x | | 0.195 | 0.235 | 0.275 | - | |
| | | y | | 0.680 | 0.720 | 0.760 | - | |
| | Blue | x | | 0.113 | 0.143 | 0.173 | - | |
| | | y | | 0.015 | 0.045 | 0.075 | - | |
| Color Gamut | | | vs. NTSC (CIE 1931) | 90 | 105 | - | % | |
| Viewing angle | Left | θL | CR≥10 | 75 | 80 | - | Deg. | Note 3 |
| | Right | θR | | 75 | 80 | | Deg. | Note 3 |
| | Top | ϕT | | 75 | 80 | | Deg. | Note 3 |
| | Bottom | ϕB | | 75 | 80 | | Deg. | Note 3 |
| Color Shift | | | White @ 30 degree | | | 6 | JNCD | Note 4 |
| Flicker | | | | | | -30 | dB | Note 5 |
| Cross Talk | | | | | | 1.7 | % | Note 6 |
| Polarization Direction | | | PdF | | 135 | | Deg. | Note 7 |
| OLED Life Time | | | With a Full-white image, lighting on with brightness of 350 nits for 120 hrs. | T94>=120h | | | | |
| Response time | | | | | | 2 | ms | Note 8 |

Note 1: Brightness Uniformity

- Environmental conditions: Temp. 25℃±3℃, 65±20%RH, Dark Room.

- Measurement equipment: CS2000 or similar equipment.
- The brightness uniformity is calculated by using following formula:
  Brightness uniformity = Bri.(Min.) / Bri.(Max.) × 100%
  Bri.(Min.) = Minimum brightness measured in 9 measuring spots.
  Bri.(Max.) = Maximum brightness measured in 9 measuring spots.
- Illustration of 9 measuring spots as follows




Note 2: Contrast Ratio

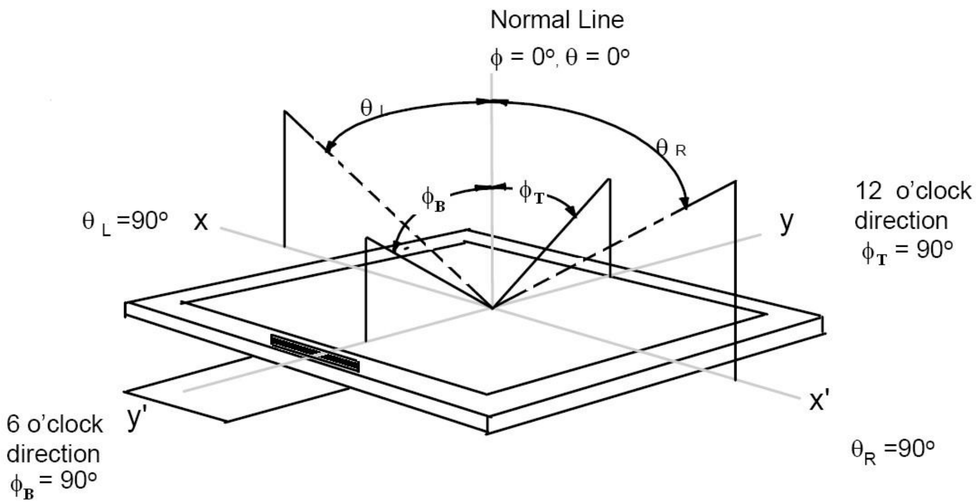
Dark Room C.R=$L_W/L_B$

$L_W$: Full white brightness of display center P0;

$L_B$ : Full black brightness of display center P0.

Note 3: Viewing Angle

Refer to the figure below marked by θ and ϕ.




Note 4: Color Shift JNCD

- For JNCD measure, fix on white pattern, on the condition θ = 0°, ϕ= 0°, a color coordinate ($u'_1$, $v'_1$) can be obtained and another color coordinate ($u'_2$, $v'_2$) can be obtained on $\theta_L$=30°.
- Delta u'v' = square root (($u'_2$-$u'_1$)^2+($v'_2$-$v'_1$)^2), and JNCD stands for "Just Noticeable Color Difference". For the (u', v') color space 1 JNCD = 0.004 Delta u'v', For example, color shift less than 2 JNCD means Delta u'v' < 0.008.

Note 5: Flicker

- Measurement equipment: CA-210 or similar equipment.
- Measuring temperature: Ta = 25 °C.
- Test method: JEITA method.
- Test pattern: Refer to below (Test pattern should be full-fill of display

screen).

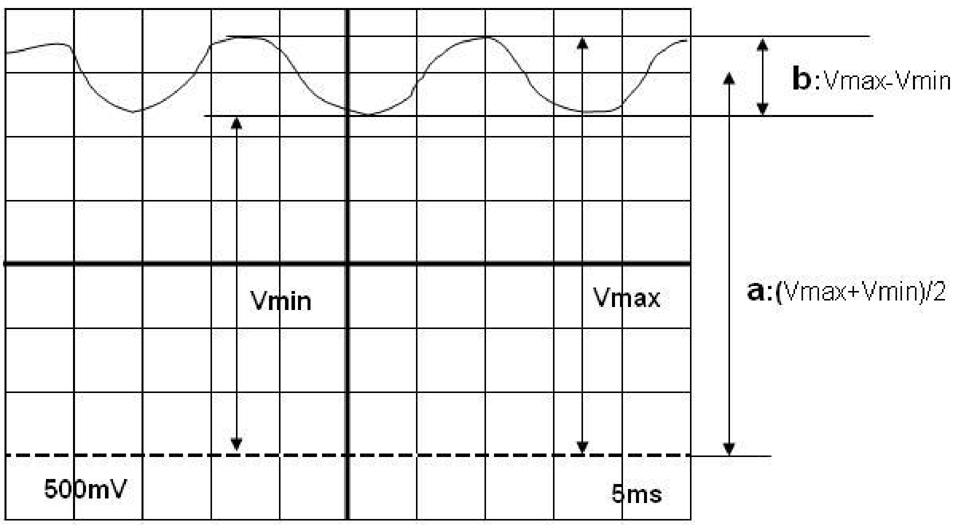
- The point should be marked is, the background of Flicker test pattern – "gray" is defined as middle gray scale. For example, RGB 24 bit "gray" is defined as below:

| R7 | R6 | R5 | R4 | R3 | R2 | R1 | R0 | G7 | G6 | G5 | G4 | G3 | G2 | G1 | G0 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

- Frame frequency requirement before test: The display panel must be tuned to more than 60 Hz before measurement.

- If the intensity level of the display changes as Figure below, it is considered that AC component (b) overlaps on the DC component (a). With the contrast method, the ratio of AC component to DC component is defined as the flicker amount.

- AC component (b) is defined as $V_{max}$ - $V_{min}$ and DC component (a) as ($V_{max}$ + $V_{min}$)/2, and the flicker amount is calculated by the following formula:
  Flicker amount = AC component / DC component = b/a
  $= (V_{max} - V_{min}) / [(V_{max} + V_{min})/2] \times 100\%$



Note 6: Crosstalk

- Measurement equipment: CS2000 or similar equipment.
- The background of crosstalk test pattern 'gray' is defined as middle gray scale. For example, RGB 24 bit 'gray' is defined as below'

| R7 | R6 | R5 | R4 | R3 | R2 | R1 | R0 | G7 | G6 | G5 | G4 | G3 | G2 | G1 | G0 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

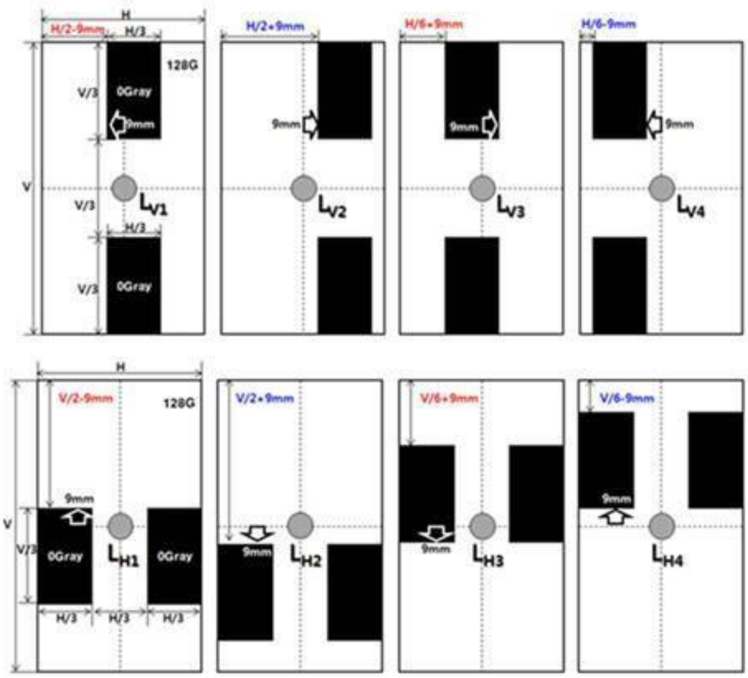
- Test pattern follows the picture below, the background is middle gray and with two black rectangle parts, each one is 1/9 of the AA size.
- Calculate the crosstalk (V) and crosstalk (H) with the formula below:

$$Crosstalk\ (V) = \max\left(\left|\frac{L_{V1} - L_{V2}}{L_{V2}}\right| \times 100, \left|\frac{L_{V3} - L_{V4}}{L_{V4}}\right| \times 100\right)$$

$$Crosstalk\ (H) = \max\left(\left|\frac{L_{H1} - L_{H2}}{L_{H2}}\right| \times 100, \left|\frac{L_{H3} - L_{H4}}{L_{H4}}\right| \times 100\right)$$

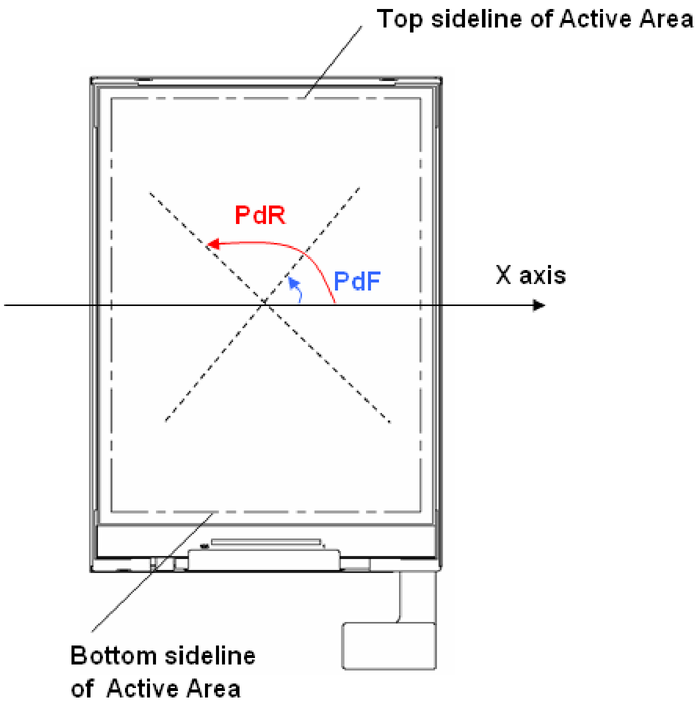
- The final crosstalk value is the maximum one between Crosstalk (V) and Crosstalk (H).

**Pattern : 2/9 area is Black Box , (Background: 128gray)**



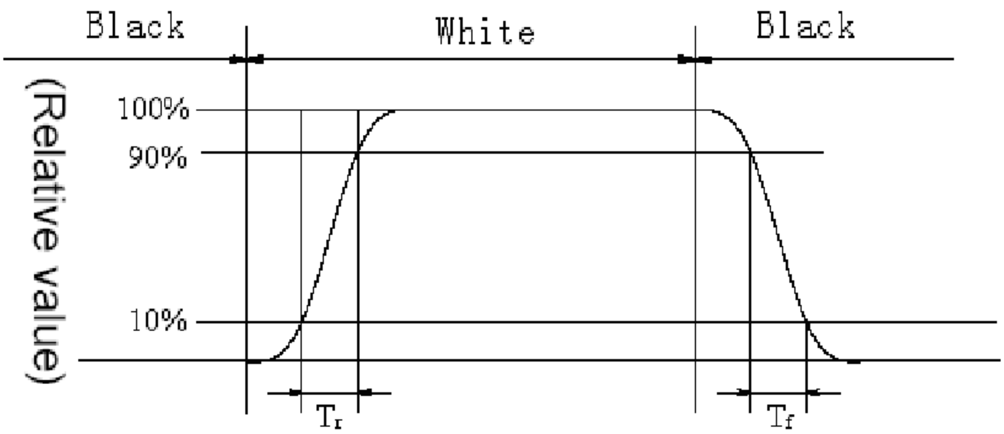

Note 7: Polarization Direction Definition

- Viewing direction is normal user viewing direction which is vertical to the display surface.
- The X axis is defined as parallel to Top and Bottom sidelines of the Active Area.
- PdF which is marked in blue arrow in the figure below is polarization degree.
- The polarization degree parameter is indicated in range of 0 deg. to 180 deg. according to above definition.




Note 8: Definition of Response Time

The output signals of photo detector are measured when the input signals are changed from 'black' to 'white' (voltage falling time) and from 'white' to 'black' (voltage rising time), respectively. The response time is defined as the time interval between 10% and 90% of the amplitudes, as shown in the figure below:




- Response time of gray to gray

Measurement equipment: CS2000 or similar equipment.

Test method: 8 grays L0 to L7 are defined, the gray level of which are 0, 36, 73, 109, 146, 182, 219, and 255. The output signals of photo detector are measured when the input signals are changed from 'Lx' to 'Ly', [x, y] = [0, 7]. The response time is defined as the time interval between 10% and 90% of the amplitudes. The result of the test can be noted as below:

| | L0 | L1 | L2 | L3 | L4 | L5 | L6 | L7 |
|---|---|---|---|---|---|---|---|---|
| L0 | | | | | | | | |
| L1 | | | | | | | | |
| L2 | | | | | | | | |
| L3 | | | | | | | | |
| L4 | | | | | | | | |
| L5 | | | | | | | | |
| L6 | | | | | | | | |
| L7 | | | | | | | | |

# 7 Reliability

## 7.1 Environmental Test

| No | Item | Conditions | Note |
|---|---|---|---|
| 1 | High Temperature Storage 5pcs | 85℃ / 128 hours | Base on FOG or Full-MDL Base |
| 2 | Low Temperature Storage 5pcs | -40℃/128hours | on FOG or Full-MDL |
| 3 | High Temperature Operation 5pcs | 70℃ / 128 hours | Base on FOG or Full-MDL Base |
| 4 | High Temperature and Humidity Operation 5pcs | 60℃ / 90% RH 128 hours | on FOG or Full-MDL |
| 5 | Low Temperature Operation 5pcs | -20℃ / 128 hours | Base on FOG or Full-MDL |
| 6 | Thermal Shock Storage | -40℃ ~85℃ 30min, | Base on FOG or |

| | | | |
|---|---|---|---|
| | 5pcs | change time < 5min, 30 cycles | Full-MDL |
| 7 | Air discharge<br>5pcs | ±8 KV,150PF/330Ω<br>5 Points (1 Center / 4 Corners),<br>Each 2 times | Base on Full-MDL<br>Note 10 |
| 8 | Contact discharge<br>5pcs | ±4 KV, 150PF/330Ω<br>5 Points (1 Center / 4 Corners)<br>Each 2 times | Base on Full-MDL<br>Note 10 |

**Note 10:Class C will be executed unless otherwise specified :**

| | Class | Performance |
|---|---|---|
| ESD Criterion | A | All functions perform as designed during and after exposure to interference |
| | B | Temporary degradation or less of performance which is self-recoverable |
| | C | Degradation or less of performance which requires operator intervention or system reset to recover |
| | D | Degradation or less of function which is not recoverable |

## 7.2 Mechanical

| No | Item | Main spec | Note |
|---|---|---|---|
| 1 | 4PB | >80MPa | |
| 2 | Drop Test | Note11 | Package |
| 3 | Sinusoidal Vibration Test | Note12 | Package |

**Note 11 Drop Test**

试验前根据包装件重量从表 1 中选择对应的跌落高度进行跌落试验。

对包装件进行 1 个角、3 条棱和 6 个面的跌落试验，具体角、棱和面跌落对象及跌落顺序表 1。

| 重量 M（Kg） | 跌落高度（mm） | 试验次数 |
|---|---|---|

| | | |
|---|---|---|
| M＜10 | 1000 | 每个指定<br>的面、角和棱各<br>跌落 1 次 |
| 10≤M＜15 | 1000 | |
| 15≤M＜20 | 800 | |
| 20≤M＜30 | 600 | |
| 30≤M＜40 | 500 | |
| 40≤M＜50 | 400 | |
| 50≤M＜100 | 300 | |

表 1　自由跌落试验参数表

| 试验顺序 | 跌落对象 | 确定跌落对象方法 |
|---|---|---|
| 1 | 1＃角 | 最薄弱的一个底角；不能确定的话，选择角 2-3-5 |
| 2 | 1＃棱 | 构成最薄弱底角的三条棱之中最短的一条棱 |
| 3 | 2＃棱 | 构成最薄弱底角的三条棱之中倒数第二短的一条棱 |
| 4 | 3＃棱 | 构成最薄弱底角的三条棱之中最长的一条棱 |
| 5 | 1＃面 | 包装件外表面中面积最小的一个 |
| 6 | 2＃面 | 与 1＃面正对的面 |
| 7 | 3＃面 | 包装件外表面中面积中等的一个 |
| 8 | 4＃面 | 与 3＃面正对的面 |
| 9 | 5＃面 | 包装件外表面中面积最大的一个 |
| 10 | 6＃面 | 与 5＃面正对的面 |

表 2　自由跌落试验顺序表

测试适用范围：外箱，最少数量 1PCS 外箱。包装内顶层和底层 Tray 盘必须装满良品，中间 Tray 盘可以装不良品配重。

合格判据

1）中箱允许不影响运输保护的破损（破损长度<5cm），允许适当的变形。不允许机能上的损伤

2）包装的缓冲材料应无变形、允许小量的不可恢复的压痕、折痕和破裂。(破损长度<3cm)，但需要记录失效现象并拍照。

3）Tray 盘不可以有开裂、折痕和不开恢复变形问题。

4）试验后顶层和底层样品机械功能和电气功能正常。

**Note 12 Vibration Test**

试验时按照表 3、表 4 的参数进行试验设置：

| 试验顺序 | 试验轴向 | 试验时间（min） | 备注① |
| --- | --- | --- | --- |
| 1 | Z 轴向 | 30 | 面 3 放置工作台上，装夹方式是压紧 |
| 2 | Y 轴向 | 30 | 面 4 放置工作台上，装夹方式是压紧 |
| 3 | X 轴向 | 30 | 面 6 放置工作台上，装夹方式是压紧 |

表 3 随机振动试验参数表

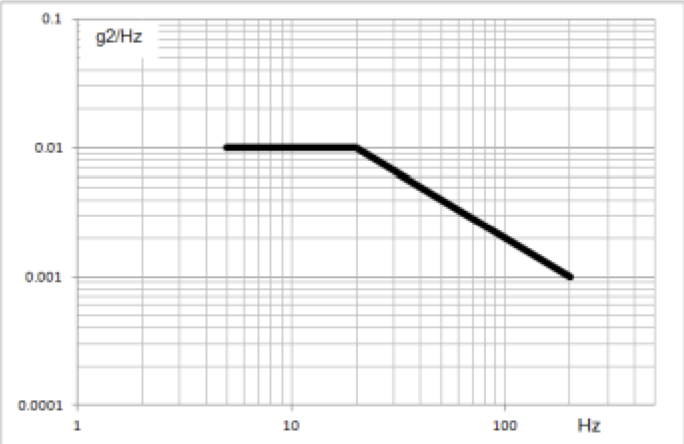
若产品有明确的放置面，则要求放置面始终放在振台面上。即 Y 轴向和 X 轴向测试放在振动台的水平滑台上。

| 频率（HZ） | PSD（功率谱密度，g2/Hz） |
| --- | --- |
| 5 | 0.01 |
| 20 | 0.01 |
| 200 | 0.001 |

| | |
|---|---|
| 均方根加速度（Grms） | 0.781 |

表 4　随机振动断点设置表

根据上表设置的随机振动频谱曲线（PSD VS Hz）应符合下图：




测试适用范围：外箱，最少数量 1PCS　外箱。包装内顶层和底层 Tray 盘必须装满良品，中间 Tray 盘可以装不良品配重。

合格判据

1）中箱允许不影响运输保护的破损（破损长度<5cm），允许适当的变形。不允许机能上的损伤

2）包装的缓冲材料应无变形、允许小量的不可恢复的压痕、折痕和破裂。(破损长度<3cm)，但需要记录失效现象并拍照。

3）Tray　盘不可以有开裂、折痕和不开恢复变形问题。

4）试验后顶层和低层样品机械功能和电气功能正常。

# 8 Handling Precautions

1) When cleaning ITO pad, avoid using hard and abrasive material or corrosive solution
2) Keep module away from direct sunlight or fluorescent light, and keep it at room temperature and humidity
3) Strong impact & pressure on module and packing is prohibited
4) Following normal power on/off sequence is necessary for preventing abnormal display or permanent damage to display
5) Optimal contrast ratio under ideal voltage is AMOLED module′ s characteristic, hence it is recommended a voltage control function available.
6) Image sticking may occur if an image displays for an extended period of time
7) When interfered by system's overall mechanical design, an abnormal display may occur
8) After considering emitting energy, you should plan your design to satisfy EMI standards.
9) Host side should place a surge-prevent circuit at power trace (ie: VCI, Vddi) to protect AMOLED module.

## 9 LCM Dimension Drawing

### 9.1 LCM Module Outline